

SAHIL TOMAR

Ghaziabad, India ·  human-in-loop.dev

 (+91) 94578 18390

 dev.sahil.tomar@gmail.com

 linkedin.com/in/dev-s-t

 github.com/dev-S-t

Summary

AI Solutions Engineer leading the Voice AI vertical (VOAG) at TechieMaya — production real-time voice infrastructure on LiveKit/WebRTC and SIP telephony, running **1,000+ calls/day at sub-200ms p95 latency**.

Owns systems end to end: architecture, deployment *inside* client infrastructure under NAT and compliance constraints, and production monitoring. Builds agentic systems with Google ADK, RAG over private data, and generative media pipelines.

Work Experience

TechieMaya

Apr. 2025 – Present

AI Solutions Engineer (Intern → Full-Time Contract)

Remote (Dubai, UAE)

– VOAG — Enterprise Voice AI SaaS

Mr. LADs

- **Lead the Voice AI vertical:** scaled to **1,000+ calls/day at sub-200ms p95 latency**; replaced persistent WebSockets with an *async polling layer* waking Cloud Run workers on demand, cutting infrastructure cost **30%**.
 - **Hybrid cloud SIP:** engineered a UAE telephony bridge (SIM → modem → SIP → LiveKit) deployed *inside a client's own VM behind a strict NAT gateway* for ultra-low-latency local calling; self-hosted LiveKit across India and UAE regions with tenant-aware call routing.
 - **Agentic workflows:** built a tenant-aware tool registry over OAuth 2.0 enabling live calendar scheduling, omnichannel messaging, and human handoff mid-call; added *fire-and-forget RAG* so agents reference long documents without conversational dead air.
 - **Multi-provider & delivery:** integrated Gemini Live, Sarvam, and Ultravox for cost/language routing; containerised all workers with **GitHub Actions** CI/CD, plus a playground for testing agents against the production pipeline.
- #### – WhatsApp Dispatch Automation — B2B, luxury ground transport (Dubai)
- Built a Go gateway (*whatsmeow*) for group-chat messaging unsupported by Meta's official API, paired with a **Google ADK multi-agent system** — orchestrator, booking agent, and support agent — resolving flight, maps, fleet-availability and fare data via tools, and asking targeted follow-ups on incomplete requests.
 - Cut booking turnaround **from ~30 minutes to under 1 minute**; extended the same gateway for a second client with message-monitoring and multi-mapping forwarding services across 1,000+ daily community messages.
- #### – MAGE — Multi-Agent Media Generation Engine
- Hierarchical agent pipeline (creative director → scriptwriter → reference selector → generator → reviewer) producing on-brand video ads from a **Playwright**-scraped brand profile, using Google Veo with frame-carry continuity and FFmpeg assembly.
 - Added Reddit trend intelligence for creative direction and per-session API cost accounting via Python `contextvars` across concurrent async workers.
- #### – Privacy-First On-Premise RAG (A2A Protocol)
- Inverted RAG topology for compliance-sensitive clients — the reasoning agent deploys *onto client infrastructure* over an **Agent-to-Agent protocol**, queryable by any external agent with **zero data exfiltration**; hybrid dense-vector and keyword index for mixed document types.

Quantashift Consultancy Services (contracted to MGS Technology)

Jan. 2026 – Mar. 2026

Freelance Voice AI Consultant

Remote (Pune, India)

– Hireups — Real-Time AI Interviewer with Photorealistic Video Avatar

Hireups

- Brought in after the internal team was blocked for months on LiveKit real-time *video*; delivered a **live photorealistic avatar interviewer** on Gemini's realtime audio-native model with sub-second, interruptible multilingual dialogue (English/Arabic in production).
- Shipped **three production deployments** — GCP VM, Cloud Run (non-trivial: LiveKit workers require persistent connections), and the client's private server, which needed a *custom LiveKit build* to satisfy ZeroSSL on an untrusted IP range.
- Rearchitected a failing WebRTC monolith into scalable GCP microservices; added a **multi-LLM failover mesh** (Gemini/Groq/Ultravox) for uninterrupted sessions, and moved proctoring inference in-browser (TensorFlow.js, MediaPipe, COCO-SSD) to eliminate server video compute.

Infosys Springboard

Mar. 2024 – Jul. 2024

Machine Learning Intern

Remote (GitHub)

- Built a hybrid fraud-detection pipeline on a highly imbalanced **400+ feature** dataset (ARIMA/Prophet + XGBoost), achieving **96% fraud recall** with an optimised F1-score.

Publication

A Demand-Driven Software Approach with Dynamic Micro-Expiry and Just-in-Time Processing to Reduce Platelet Wastage in Blood Banks

- *IEEE IC2PCT 2026*, pp. 978–983

[IEEE Xplore](#)

- Co-author — led problem formulation, forecasting-model development (SARIMA/XGBoost; SARIMA lowest MAE at 5.85), and simulation. Reduced simulated wastage **11.2%** → **2.5%** while holding **99.1%** fulfillment; validated over 30 iterations (paired t-tests, $p < 1.22 \times 10^{-12}$).

Projects

AnyAssist — Multi-Tenant RAG Platform — *GCP, LiteLLM, LightRAG, FastAPI, Google ADK*

[AnyAssist](#)

- Self-service “chat with your data” SaaS with strict per-tenant vector isolation; LiteLLM gateway handles cross-provider routing, rate limiting, and per-tenant cost tracking.

UniBias — Live Attention Tracker — *Vision Transformers, FastAPI, OpenCV*

[UniBias](#)

- Privacy-first tracker analysing periodic webcam and screen frames to detect distraction and trigger alerts, with no recording.

Technical Skills

Languages & Infra: Python (AsyncIO), Go, SQL, Docker, GCP (Cloud Run, Compute Engine, Vertex AI), DigitalOcean
Voice & Real-Time: LiveKit (WebRTC), SIP/VoIP bridging, WebSockets, Gemini Live API, Sarvam, Ultravox, streaming STT/TTS

AI & Agents: Google ADK, LangGraph, LiteLLM, LightRAG, Multi-Agent Systems, A2A Protocol, RAG, Vector DBs

Generative Media: Google Veo, Imagen, Cartesia TTS, FFmpeg, Playwright

Backend & DevOps: FastAPI, REST APIs, OAuth 2.0, Redis, NGINX, GitHub Actions (CI/CD), Fail2Ban

Education & Extracurricular

Ajay Kumar Garg Engineering College (AKGEC)

Sept. 2022 – Jun. 2026

B.Tech in Computer Science · CGPA: 8.0

Ghaziabad, India

- Coordinator, Cloud Computing Cell:** organised ML and cloud workshops for 200+ students.
- Coordinator, Centre of Metaverse:** led “Rescue X”, a VR first-responder platform, to **Top 5** at the National IDE Boot-camp 2025.

Certifications

- Building AI Voice Agents for Production, LiveKit / DeepLearning.AI (Sep. 2025) · Foundations of Responsible AI (Sep. 2025)
- Complete Data Science, ML, DL & NLP Bootcamp, Udemy (Jul. 2025) · Fundamentals of Deep Learning, NVIDIA DLI (Mar. 2025)